

$$\begin{aligned}\therefore 80M + 100D &= 80M + 100 \times \frac{M}{5} \\ &= 80M + 20M = 100M.\end{aligned}$$

So, Ravi can read 100 pages of Engineering Mathematics.

- 509.** Since Rohan paid ₹ 20 and because of lack of change, the clerk gave him stamps worth ₹ 3, it can be concluded that the total value of the stamps that he wanted to buy is ₹ 17. Since he ordered initially a minimum of 2 stamps of each denomination, if he bought exactly 2 stamps each, his total value is ₹ $[2(5 + 2 + 1)] = ₹ 16$. To make it ₹ 17, he must have bought 1 one-rupee stamp more.

So, total number of stamps ordered = $(2 + 2 + 3) = 7$.

Since the clerk gave him 3 more, so total number of stamps bought = $(7 + 3) = 10$.

- 510.** Suppose Aditya bought x gel pens and y fountain pens. Then.

$$300x + 400y = 3600 \Rightarrow 3x + 4y = 36 \quad \dots(i)$$

Also, $(300x + 400y) - (400x + 300y) = 150$ or 200

$$\Rightarrow 100y - 100x = 150 \text{ or } 200$$

$$\Rightarrow y - x = 1.5 \text{ or } y - x = 2.$$

If $y - x = 1.5$, we don't get integral values for x and y .

$$\therefore y - x = 2 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 4$, $y = 6$.

Hence, total number of pens bought = $4 + 6 = 10$.

Questions 511 to 513

Flowers left after first place of worship = $2x - y$.

Flowers left after second place of worship = $2(2x - y) - y$
 $= 4x - 3y$.

Flowers left after third place of worship = $2(4x - 3y) - y$
 $= 8x - 7y$.

Flowers left after fourth place of worship = $2(8x - 7y) - y$
 $= 16x - 15y$.

$$\therefore 16x - 15y = 0 \Rightarrow 16x = 15y.$$

- 511.** If $x = 30$, then $y = \frac{16 \times 30}{15} = 32$.

- 512.** For minimum y , we find the L.C.M. of 16 and 15 which is 240.

$$\therefore 16x = 15y = 240 \Rightarrow x = 15, y = 16.$$

- 513.** Clearly, required number = minimum value of $x = 15$.

- 514.** Suppose the thief stole x diamonds.

$$\text{Diamonds given to first watchman} = \frac{x}{2} + 2 = \frac{x+4}{2}.$$

$$\text{Remaining diamonds} = x - \left(\frac{x+4}{2}\right) = \frac{x-4}{2}.$$

$$\text{Diamonds given to second watchman} = \frac{x-4}{4} + 2 = \frac{x+4}{4}.$$

$$\text{Remaining diamonds} = \left(\frac{x-4}{2}\right) - \left(\frac{x+4}{4}\right) = \frac{x-12}{4}.$$

$$\text{Diamonds given to third watchman} = \frac{x-12}{8} + 2 = \frac{x+4}{8}.$$

$$\text{Remaining diamonds} = \left(\frac{x-12}{4}\right) - \left(\frac{x+4}{8}\right) = \frac{x-28}{8}.$$

$$\therefore \frac{x-28}{8} = 1 \text{ or } x-28=8 \text{ or } x=36.$$

Another Method: The thief escaped with 1 diamond.

Before third watchman, he had $(1 + 2) \times 2 = 6$ diamonds.

Before second watchman, he had $(6 + 2) \times 2 = 16$ diamonds.

Before first watchman, he had $(16 + 2) \times 2 = 36$ diamonds.

Thus, he stole 36 diamonds originally.

- 515.** Let there be x mints in the bowl originally.

Then, mints left after Divya had taken from the bowl

$$= \left(\frac{2}{3}x + 4\right).$$

Mints left after Reema had taken from the bowl

$$= \frac{3}{4} \left(\frac{2}{3}x + 4\right) + 3 = \left(\frac{x}{2} + 6\right).$$

Mints left after Shweta had taken from the bowl

$$= \frac{1}{2} \left(\frac{x}{2} + 6\right) + 2 = \left(\frac{x}{4} + 5\right).$$

$$\therefore \frac{x}{4} + 5 = 17 \Rightarrow \frac{x}{4} = 12 \Rightarrow x = 48.$$

- 516.** Let the cost of one gel-pen, one ball-point pen and one marker pen be ₹ x , ₹ y and ₹ z respectively.

$$\text{Then, } 4x + 8y + z = 185 \quad \dots(i)$$

$$7x + 15y + z = 315 \quad \dots(ii)$$

$$\text{Subtracting (i) from (ii), we get: } 3x + 7y = 130 \quad \dots(iii)$$

Multiplying (iii) by 2 and subtracting from (ii), we get:
 $x + y + z = 55$

Hence, required cost = ₹ 55.

- 517.** Let the number of byes, wides and runs be x , y and z respectively.

Then, $x + y + z = 232$. Also, $z = 26y$ and $x = y + 8$.
 $\dots(i)$

$$\therefore (y + 8) + y + 26y = 232 \Leftrightarrow 28y = 224$$

$$\Leftrightarrow y = 8. \text{ So, } z = 26y = 208.$$

Let the number of runs scored by Ram and Shyam be $6a$ and $7a$ respectively.

$$\text{Then, } 6a + 7a = 208 \Leftrightarrow 13a = 208 \Leftrightarrow a = 16.$$

Hence, runs scored by Ram = $6a = 6 \times 16 = 96$.

- 518.** Let there be n rows of balls in the equilateral triangle.

$$\text{So, total number of balls} = 1 + 2 + \dots + n = n = \frac{n(n+1)}{2}.$$

Since each row in the square has 8 balls less than that in the equilateral triangle, so the square has $(n - 8)$ rows with $(n - 8)$ balls in each row.

So, total number of balls = $(n - 8)^2$.

$$\therefore \frac{n(n+1)}{2} + 669 = (n-8)^2$$

$$\Rightarrow n^2 + n + 1338 = 2(n^2 + 64 - 16n)$$

$$\Rightarrow n^2 - 33n - 1210 = 0$$

$$\Rightarrow n^2 - 55n + 22n - 1210 = 0$$

$$\Rightarrow (n - 55)(n + 22) = 0 \Rightarrow n = 55 \quad [\because n \neq -22]$$

So, initial number of balls

$$= \frac{n(n+1)}{2} = \frac{55 \times 56}{2} = 55 \times 28 = 1540.$$

Questions 519 and 520

Let the number of men in the beginning be m .

Suppose each man shifts c crates on the first day. Then,

$$\begin{aligned}mc + (m + 6)(c - 5) + (m + 12)(c - 10) + (m + 18)(c - 15) + (m + 24)(c - 20) &= 545\end{aligned}$$